Knot Theory

Session 1

Recap

1. Knot: an embedding of a circle in $\mathbb{R}^3$
2. Link: an embedding of two or more circles in $\mathbb{R}^3$
3. Ambient Isotropy: equivalent under Reidemeister moves
4. Reidemeister move
    i. twist
    ii. poke: 2 parallel strands, bend one under another
    iii. slide (under a crossing)

left-handed: over strand right-down to left-up

Building knots

1. Mirror knots: reflect, $m(K)$
2. Amphichiral: $K \neq m(K)$, otherwise chiral
3. Oriented knots
    i. over / under-crossing along direction of arrow
4. Reverse knots
    i. Reverse knot K^{-1}: change orientation
    ii. Invertibility: K equivalent to K^{-1}
5. Knot composition $\#$
    i. Def
        a. cut 2 oriented knots
        b. composed with corresponding orientation
        c. no new crossings introduced
        d. prime: cannot be composed of non-trivial oriented knots
    ii. Properties (integer multiplication)
        a. commutative
        b. associative
        c. identity
        d. NO inverses: unknot is prime

    e. Prime factorisation of knots
6. Alternating knots: over & under-crossings alternates as traveling along a knot in a fixed direction
    i. e.g. trefoil going twice
    ii. cannot twist or poke preserving alternative

Notations of knots

1. Gauss notation (not useful)
    i. give a number for each crossing
    ii. start from the top, anti-clockwise, + over-crossing, - under-crossing
2. Dowker-Thistlethwaites (drow the graph with numbers)
    i. double marking crossings along the knot, write down even numbers corresponds to 1357...
    ii. $+$ if even is over-crossing
    iii. all + if alternating
3. Alexander-Briggs
    i. Determine crossing numbers
    ii. Underscore knots under each crossing number
4. Conway: tangles (most useful)
    i. Tangle: a region of knot / link projection s.t. exactly 4 strands cross the boundary, denote as NW, SW, NE, SE
    ii. over-strand goes up: + (positive gradient)
    iii. Join north / south strands
    iv. Compose
        a. start from a plain region e.g. +2
        b. reflect over $y = -x$
        c. compose with another region
    v. Continued fraction: denominator contains the sum of another fraction (infinite or finite)
        a. $a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \dots}}$
        b. same tangle same continued fraction

Counting knots

1. Tait's Menage Problem
2. Crossing numbers $c(K)$: min # of crossings in any projection of K
3. Thm: reduced (no simplifications under I and II) alternating projection with n crossings, $c(K) = n$

Useful: search knot atlas